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PiHub

Matter-enabled smart Home system

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What is PiHub?

PiHub is a local IoT smart home hub built on a Raspberry Pi 4. It uses the Matter open standard to commission, monitor, and control custom ESP32-C3 firmware nodes entirely on-device, with no cloud dependency

Key Features:

- Matter protocol commissioning.
- Real-time device control via python-matter-server WebSocket.
- On-device TinyML context classification running on the ESP32-C3 Sensor Node.
- Touch-first kiosk UI running on Raspberry Pi Touchscreen.
- Full GitHub actions CI/CD Pipeline.
- Deployment repo pulling GHCR images.

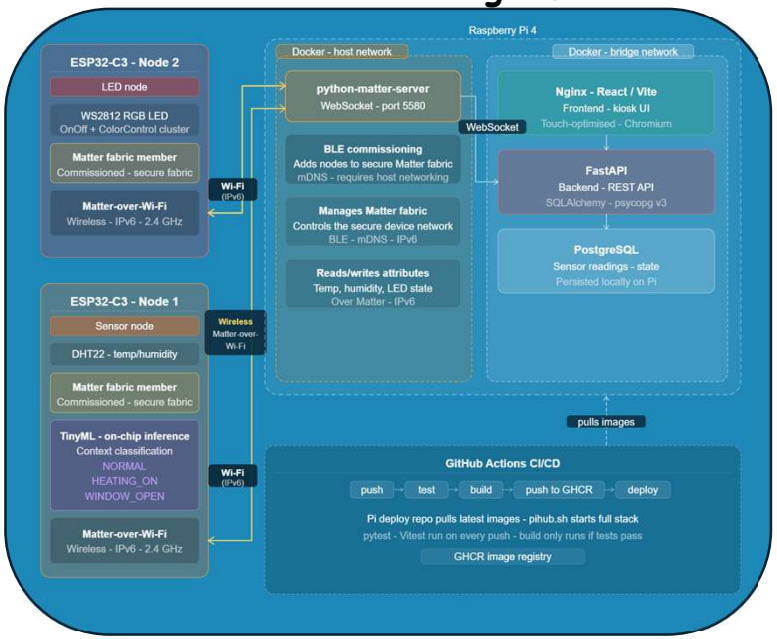
How Does it Work?

1. **Sensor Node (ESP32-C3 + DHT22)** Reads temperature & humidity. An int8-quantised **TinyML** autoencoder runs inference on-chip and reports a context label via the MIN_MEASURED_VALUE cluster.

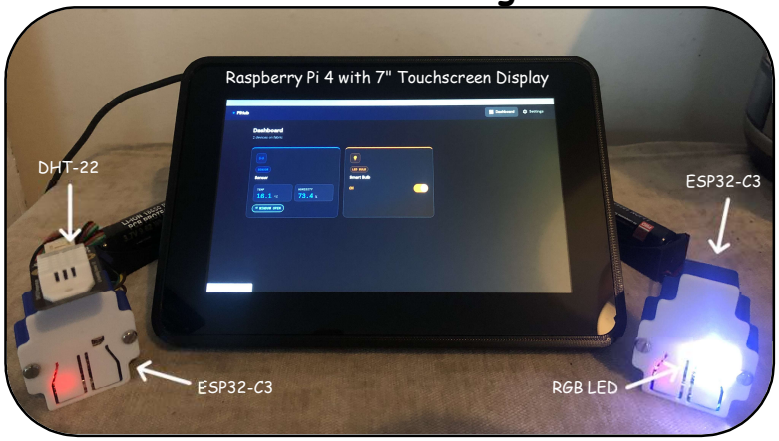
2. **LED Node (ESP32-C3 + WS2812)** Receives on/off, brightness, and CIE XY colour commands over Matter. Colour is converted through a full sRGB gamma linearisation pipeline for accurate output.

3. **Hub (Raspberry Pi 4)** Runs python-matter-server (WebSocket), FastAPI backend, PostgreSQL database, and React/Vite frontend all in Docker. Live sensor data is served via cached /live endpoints polled by the UI & LED state is controlled via the Hub.

Architecture Diagram



Hardware Design



Technologies Used

PostgreSQL, React, FastAPI, VITE, TensorFlow, NGINX, matter, GitHub, docker

Technical Highlights

1. **Matter** is an open-source smart home connectivity standard backed by Apple, Google, & Amazon. It allows devices from different brands to communicate over a local network without relying on the cloud, making smart home systems more private.

2. **TinyML** enables quantised ML models to run on microcontrollers. In PiHub, a lightweight model on the ESP32-C3 performs real-time inference on sensor data to classify NORMAL, HEATING_ON, & WINDOW_OPEN states entirely on-device - eliminating latency & cloud dependency.

3. The **GitHub Actions** pipeline builds and publishes container images to GHCR. The Raspberry Pi pulls the latest images and runs two Docker stacks: a host-networked stack for python-matter-server, & a bridge-networked stack for FastAPI, PostgreSQL, and React - enabling automated deployment.

Conclusion

PiHub demonstrates a local, industry-standard smart home hub. The system successfully implements the Matter protocol, runs on-device TinyML context classification, and is deployed via a fully automated CI/CD pipeline. The result is a private IoT system that spans embedded firmware, backend services, modern UI design, and on-device ML.